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Research on interference avoidance effect of OneWeb satellite constellation's progressive pitch strategy

Ting Li¹  | Jin Jin²  | Wei Li³ | Zixuan Ren¹  | Linling Kuang²

¹School of Aerospace Engineering, Tsinghua University, Beijing, China

²Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing, China

³Satellite Frequency & Orbit Resource Research Center, The State Radio Monitoring Center, Beijing, China

Correspondence

Jin Jin, Main Building 10-101, Tsinghua University, Haidian District, Beijing 100084, China.
Email: jinjin_sat@tsinghua.edu.cn

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Summary

Due to the scarcity of spectrum resources, the non-geostationary orbit (NGSO) satellite constellations, which have achieved rapid development in recent years, have to deal with the challenge of spectral coexistence with geostationary orbit (GSO) satellite systems. OneWeb has put forward the “progressive pitch” strategy for GSO networks protection; however, the technical details are not exposed. Herein, we propose a method to obtain the exact progressive pitch parameters suitable for OneWeb system in terms of the technical route of this strategy. Constrained by GSO satellite system's interference protection criterion, an optimization problem is formulated by mathematical modeling of the spectral coexistence scenario between OneWeb system and a GSO satellite system. This optimization problem is resolved by a set of transformations, and the progressive pitch parameters with less loss of the system service capability are obtained. In addition, numerical simulation results verify the effectiveness of the progressive pitch strategy with our obtained parameters for interference avoidance in most parts of the world, but service blind areas of OneWeb system may appear in some time. Finally, we briefly analyze the causes and elimination ways of the blind areas.

KEYWORDS

interference analysis, interference avoidance, OneWeb satellite constellation, progressive pitch

1 | INTRODUCTION

The past few years have witnessed the explosive growth in space information and data, which makes the demand for broadband satellite communication networks increasingly urgent. Meanwhile, the radio spectrum resource used for space services has become continuously scarce. Spectrum sharing is adopted as one of the key technologies to enhance frequency utilization efficiency,^{1–6} which also causes that the cofrequency interference between different communication systems needs to be solved.^{7–10} In addition, as the geostationary orbit (GSO) is getting crowded, non-geostationary orbit (NGSO) satellite constellations have achieved rapid development, such as O3b, Starlink, OneWeb, and so on,^{11–14} providing global broadband services. With respect to the coexistence of multiple GSO and NGSO satellite networks in the future, more and more complex potential interference situations will appear. Therefore, interference analysis and avoidance have become indispensable in the construction NGSO constellation networks.

In order to avoid the interference between GSO and NGSO satellite networks, the International Telecommunication Union (ITU) has proposed a series of mitigation strategies in its recommendations, including spatial isolation, satellite diversity, transmission beam turning-off, power reduction, etc.^{15,16} Moreover, a number of academic studies have focused on the interference analysis and avoidance issue. Fortes et al. introduced an analytical method to quickly estimate interference involving NGSO satellite constellations.¹⁷ Park et al.

established the simulation scene of cofrequency interference from NGSO satellite system to GSO satellite system and analyzed the influence of orbital angle intervals and the satellite constellation scale on the communication bit error rate.¹⁸ A link angle probabilistic analysis method for NGSO constellation interference analysis was proposed by Jin et al.¹⁹ Sharma et al. put forward an adaptive power control technique to mitigate in-line interference for spectral coexistence of GSO and NGSO satellites,²⁰ and Zhang et al. achieved the coexistence of low earth orbit (LEO) and GSO satellite systems, by optimizing direction normal of phased array antennas to change the access elevation angles of ground users.²¹

Besides, in the field of engineering, the NGSO satellite communication systems normally take reasonable measures to achieve the cofrequency coexistence with GSO satellite networks, based on their own system characteristics. For instance, Iridium NEXT has taken full advantage of its leading in satellite launch and gained the priority in the Ka-band frequency spectrum. Thus, GSO networks must fully protect Iridium's operations, and frequency avoidance is the only workable method identified.²² As latter NGSO satellite networks, O3b and OneWeb shall not cause harmful cofrequency interference to GSO satellite networks. O3b network has well utilized its zero-inclined orbital characteristics and altitude difference from the GSO. When ground users access to O3b satellites, there is an inherent angular separation from GSO space–earth link. Therefore, potential interference only exists with GSO in narrow range of equatorial latitudes, which can be mitigated by reducing transmission power.^{12,23} As for OneWeb system, it incorporates the characteristics of its own constellation configurations into the above-mentioned basic technique “spatial isolation”, which is highly distinctive among the NGSO satellite networks. In this paper, we take the OneWeb system as our research object.

OneWeb has proposed a patented technology named progressive pitch to seek the protection for GSO networks.^{13,24} As seen in Figure 1, OneWeb satellite adjusts its attitude in accordance with the latitude of its subsatellite point. Satellites are tilted gradually and slightly as they approach the equator, where the beams originally serving the low and middle latitudes are adjusted to serve the users near the equator. When OneWeb satellite moves close to its ascending or descending node, several beams need to be shut off to further avoid the interference to GSO satellite networks.

OneWeb system has adopted the joint design of frequency and orbit at the beginning of constellation construction. The progressive pitch strategy not only makes good use of its orbital configuration characteristics but also well matches the system communication performance, which is flexible and causes less loss on system service capability. Nevertheless, both the patent specification and related technical reports of OneWeb system do not provide the technical details and specific parameters setting of the progressive pitch strategy, and in the academic field, there is currently no research related to this strategy.

In this paper, we conduct a study centered on the parameters determination process and the interference avoidance effect of OneWeb's progressive pitch strategy. Considering the spectral coexistence scenario between OneWeb system and a GSO satellite network, we proposed a method to obtain the progressive pitch parameters suitable for OneWeb, with the constraint of the interference protection criteria of GSO satellite networks, and the cofrequency compatibility simulation between OneWeb and GSO satellite network in the global scope is performed to verify interference avoidance effectiveness of OneWeb's progressive pitch strategy with the previous obtained parameters.

The rest of this paper is organized as follows. Section 2 establishes the mathematical system model and formulates the key problem. Section 3 introduces the process of searching for the progressive pitch strategy's parameters. In Section 4, numerical results are presented to evaluate the performance of progressive pitch strategy under the designed parameters. We briefly analyze the causes and elimination methods of blind areas for service in Section 5. Finally, overall conclusions of this paper are drawn in Section 6.

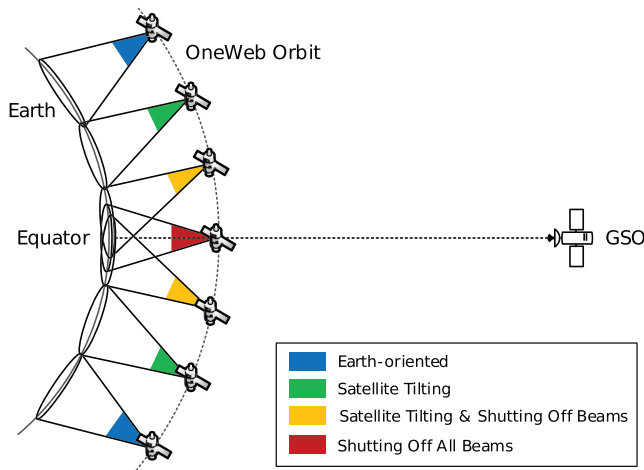


FIGURE 1 Diagram of OneWeb's progressive pitch strategy

2 | SYSTEM MODEL AND PROBLEM FORMULATION

2.1 | System model

As illustrated in Figure 2, we consider a spectral coexistence scenario of OneWeb satellite constellation and a GSO satellite network. OneWeb satellite constellation consists of 720 satellites in 18 circular orbital planes, which are all at an altitude of 1200 km and inclined at 87.9° .¹¹ Each satellite has 16 identical Ku-band nonsteerable highly elliptical user beams, with major axis in the east–west direction and minor axis in the north–south direction. As shown in Figure 3, these 16 user beams are arranged in a north–south direction single row to approximately form a large rectangular beam, whose horizontal half angle is 25° and vertical half angle is 24.5° .¹³

OneWeb satellites in different orbital planes have similar positional variations in relation to the GSO satellite and the ground. Therefore, while building the mathematical model, we take “the single OneWeb satellite” scenario as an example to carry out the following analysis. With OneWeb’s properties of the polar orbit and the large rectangular beam in the north–south direction, we consider the worst interference case, where the GSO satellite, the interfered GSO earth station, and the interfering OneWeb satellite are approximately at the same longitude. Thus, the above-mentioned coexistence scenario can be reduced in dimension, so as to simplify the complexity of analysis. Because the analytical methods of both uplink and downlink interference situations are similar, we only consider the downlink case in this paper.

As shown in Figure 4, the position vector of OneWeb satellite can be expressed as $\vec{OA} = (R_s \cos \varphi_s, R_s \sin \varphi_s)$, where φ_s represents the latitude of OneWeb’s subsatellite point and R_s is the semimajor axis of OneWeb satellite’s orbit. On the basis of the symmetry of OneWeb’s polar orbit, we take the condition that OneWeb satellite being above the Northern Hemisphere (where $\varphi_s > 0$). As for the GSO network, the position vectors of GSO satellite and GSO earth station are $\vec{OB} = (R_G, 0)$ and $\vec{OC} = (R_E \cos \varphi_{ES}, R_E \sin \varphi_{ES})$, respectively, where φ_{ES} is the latitude of the GSO earth station, R_G means the semimajor axis of GSO and R_E denotes the Earth radius. Besides, we assume that the GSO earth station is always covered by the GSO satellite and keeps its antenna’s main lobe pointing to the direction of the GSO satellite.

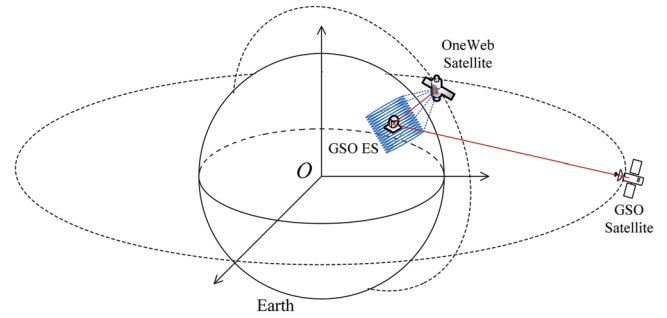


FIGURE 2 Spectral coexistence scenario of OneWeb satellite constellation network and geostationary orbit (GSO) satellite network

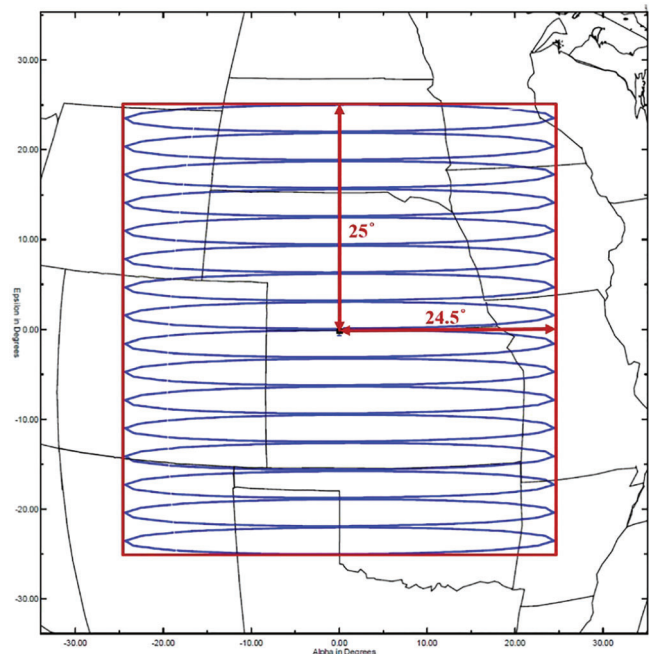


FIGURE 3 OneWeb satellite's Ku-band beam arrangement¹³

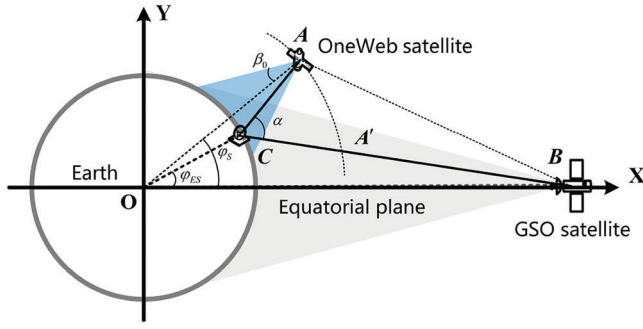


FIGURE 4 Simplified spectral coexistence scenario

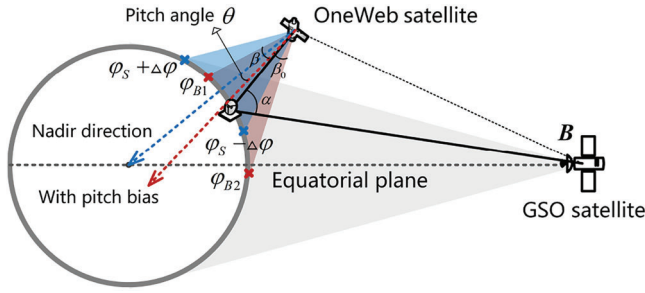


FIGURE 5 The change of beam coverage after OneWeb satellite attitude's tilting

OneWeb system adopts the progressive pitch strategy for interference mitigation. This strategy involves adjusting satellite attitude and shutting off several user beams. Next, we analyze the spatial geometric relationship while adopting this strategy. Figure 5 displays that, as OneWeb satellite is tilted to the south, its beam edge and coverage are also offset southward along with the beam center axis as a whole, and the discrimination angles of the ground positions in OneWeb's beam coverage overall increase. Moreover, shutting off highly elliptical beams can also adjust the coverage of the large rectangular beam. Let N_{OFF} represent the number of highly elliptical beams to be turned off, $\beta_{ellipse}$ be the horizontal half angle of the highly elliptical beam, and β_0 denote the original horizontal half angle of the large rectangular beam. When adjusting the beam angle of OneWeb satellite, priority is given to turning off the beam on the side away from the equatorial plane, and then the adjusted horizontal half angle of the large rectangular beam β can be expressed as

$$\beta(N_{OFF}) = \beta_0 - 2N_{OFF}\beta_{ellipse}. \quad (1)$$

As shown in Figure 5, φ_{B1} and φ_{B2} are the latitudes of the rectangular beam's edge after the beam coverage adjustment. It can be seen from the geometric relations that, φ_{B1} and φ_{B2} can be presented by N_{OFF} and the pitch angle of OneWeb satellite's attitude θ as follows:

$$\begin{cases} \varphi_{B1}(\theta, N_{OFF}) = \varphi_S + \arccos(g(\beta - \theta)) \\ \varphi_{B2}(\theta) = \varphi_S - \arccos(g(\beta_0 + \theta)) \end{cases}, \quad (2)$$

where $g(\cdot) = \frac{R_S}{R_E} \sin^2(\cdot) + \frac{1}{R_E} \sqrt{(R_E^2 - R_S^2 \sin^2(\cdot)) \cos^2(\cdot)}$. It is assumed that the GSO earth station is always within the coverage of the OneWeb satellite's beam, so

$$\varphi_{B2}(\theta) \leq \varphi_{ES} \leq \varphi_{B1}(\theta, N_{OFF}). \quad (3)$$

When the position of OneWeb satellite is determined, the discrimination angle within its beam coverage, which is formed by OneWeb satellite, GSO earth station and GSO satellite with GSO earth station being the vertex, is related to the latitude of GSO earth station φ_{ES} as follows:

$$\begin{aligned} \alpha(\varphi_{ES}) &= \arccos \left(\frac{\overrightarrow{CA} \cdot \overrightarrow{CB}}{|\overrightarrow{CA}| |\overrightarrow{CB}|} \right) \\ &= \arccos \left(\frac{R_E^2 + R_S R_G \cos \varphi_S - R_E R_G \cos \varphi_{ES} - R_E R_S \cos(\varphi_{ES} - \varphi_S)}{\sqrt{R_E^2 + R_S^2 - 2R_E R_S \cos(\varphi_{ES} - \varphi_S)} \sqrt{R_E^2 + R_G^2 - 2R_E R_G \cos \varphi_{ES}}} \right). \end{aligned} \quad (4)$$

Consider the case where downlink Ku-band frequency used by OneWeb system overlaps with that of the GSO network. When OneWeb satellite transmits signals towards the ground, it may cause harmful interference to the GSO earth stations located within its beam coverage area. The interference level at the GSO earth station's receiver can be calculated as²⁵⁻²⁷

$$I = P_T' \cdot G_{TS} \cdot G_{RG}(\alpha(\varphi_{ES})) \cdot \left(\frac{\lambda}{4\pi d_{S \rightarrow ES}} \right)^2, \quad (5)$$

where $P_T' \cdot G_{TS}$ represents the equivalent isotropically radiated power (EIRP) of the OneWeb satellite's transmitter in the GSO network's operating band. $G_{RG}(\alpha(\varphi_{ES}))$ denotes receive antenna gain of the GSO earth station in the off-axis angle $\alpha(\varphi_{ES})$ direction. In addition, let $d_{S \rightarrow ES}$ and λ be the transmission distance and wavelength of the interfering signal, respectively.

2.2 | Problem formulation

For OneWeb system, the adverse impact of adjusting satellite attitude on service capability is less than that of shutting down beams. Thus, the former is preferred. For engineering operability, OneWeb stipulates that the pitch angle of satellite should not exceed 10° .¹³ When the degree of OneWeb satellite's tilt reaches the limit but is still not enough to alleviate the interference, some highly elliptical beams need to be turned off.

In illusion to above technical features, we define the implementation cost function of the progressive pitch strategy as follows:

$$C(\theta, N_{OFF}) = \frac{\theta}{\theta_{max} + \varepsilon} + N_{OFF} + w(\theta_{max} - \theta)N_{OFF}, \quad (6)$$

where θ_{max} signifies the maximum pitch angle and $0 \leq \theta \leq \theta_{max}$. The constant ε satisfies that $\varepsilon > 0$ and $\varepsilon \rightarrow 0$, whereas the coefficient $w > 0$. In Equation (6), we can find that $\frac{\theta}{\theta_{max} + \varepsilon} \in [0, 1)$ and $N_{OFF} \in \{0, 1, \dots, N_{ellipse}\}$, which indicates that this cost function can make tilting OneWeb satellite cost less than turning off beams. Besides, the third term of the cost function in (6) represents the additional cost of the combined implementation of these two methods. As stated before, the user beams can only be shut off if the tilting degree of OneWeb satellite reaches the limit, so at least one of $\theta = \theta_{max}$ and $N_{OFF} = 0$ must be satisfied. Let $w \rightarrow +\infty$, so that the implementation cost of shutting down beams when the tilting degree of OneWeb satellite does not reach the limit ($N_{OFF} > 0$ and $\theta < \theta_{max}$) is far ahead of the first two terms, so as to avoid the occurrence of this situation. In general, the above definition conforms to the requirement of the progressive pitch strategy.

The target of this paper is to obtain the pitch angle of OneWeb satellite and the number of turned-off beams corresponding to Onweb satellite at different latitudes, according to the above-mentioned rules, that is, to achieve effective interference mitigation with minimum implementation cost. This target can be realized by solving the following optimization problem:

$$\begin{aligned} & \min_{\theta, N_{OFF}} C(\theta, N_{OFF}) \\ & \text{s.t. C1: } I_{max}(\theta, N_{OFF}) \leq I_{th}, \\ & \quad \text{C2: } 0 \leq \theta \leq \theta_{max}, \\ & \quad \text{C3: } N_{OFF} \in \{0, 1, \dots, N_{ellipse}\}. \end{aligned} \quad (7)$$

3 | SEARCH FOR THE PROGRESSIVE PITCH'S PARAMETERS

3.1 | Transformation of optimization problem

The optimization problem formulated in the previous section is in form of mixed integer, which contains the continuous variable θ and the discrete variable N_{OFF} . Considering the constraint C1, the relationship between the maximum interference level in the coverage of the OneWeb satellite's beam I_{max} and θ , N_{OFF} is not yet clear. Thus, we can not directly solve it by the usual methods suitable for the mixed-integer optimization problem. Herein, we take the constraint C1 as a breakthrough to transform this problem.

As Equation (5) in the system model expresses, the interference level I at the GSO satellite network's receiver is jointly affected by the discrimination angle α and the transmission distance of the interfering signal $d_{S \rightarrow ES}$. Therefore, our analysis is based on two assumptions:

- 1) The influence of the interfering signal propagation distance is negligible.

For the OneWeb system, $d_{S \rightarrow ES}$ is always changing due to changes in link elevation. The simulation shows that the maximum of $d_{S \rightarrow ES}$ is only

1.13 times of its minimum value, and the difference of the corresponding interference values is within 1 dB. That is, $d_{S \rightarrow ES}$ has little effect on the interference value.

2) The antenna gain G_{RG} monotonically decreases with the increase of α .

The earth stations requiring progressive pitching are located at low latitudes, so both the OneWeb and GSO satellites are accessed at high elevation. This means that the link angle between “Earth Station-OneWeb satellite” and “Earth Station-GSO satellite” is small, and generally the antenna gains will not be affected by the side lobes.

Under these assumptions, it can be seen that the interference level has decreasing property with the discrimination angle, and I reaches its maximum value when $\alpha = \alpha_{min}$. Based on the above analysis, the interference constraint C1 can be transformed into the geometrical discrimination angle constraint C1',

$$C1' : \alpha_{min}(\theta, N_{OFF}) \geq \alpha_{th}. \quad (8)$$

According to the formula (5), the threshold value of discrimination angle corresponding to the interference threshold can be defined as follows:

$$\alpha_{th} = G_{RG}^{-1} \left(\frac{I_{th} \cdot (4\pi d_{min})^2}{P_T' \cdot G_{TS} \cdot \lambda^2} \right), \quad (9)$$

where the minimum interfering signal propagation distance is $d_{min} = R_S - R_E$.

Next, we analyze the transformed constraint C1'. As shown in Equation (4), we can find that there are two main difficulties in this issue:

- (1) The derivative of α with respect to φ_{ES} is very complicated, so that its extreme point can only be derived by numerical methods such as Newton's method and it is difficult to obtain the analytical solution.
- (2) Because the definition domain of $\alpha(\varphi_{ES})$ contains variables, its monotonicity is related to θ and N_{OFF} , which can be hardly determined. We can not obtain the analytic expression of α_{min} .

To address these difficulties, we convert the constraint C1' into the equivalent form shown in Equation (10), thus avoiding obtaining the expression of $\alpha_{min}(\theta, N_{OFF})$.

$$C1'' : \alpha(\varphi_{ES}) \geq \alpha_{th}, \forall \varphi_{ES} \in [\varphi_{B2}(\theta), \varphi_{B1}(\theta, \beta)]. \quad (10)$$

The equivalent constraint C1'' means that the discrimination angle of any ground position within OneWeb's beam coverage needs to be no smaller than α_{th} . To satisfy this constraint, we need to find out the boundaries of the OneWeb satellite rectangular beam's coverage corresponding to α_{th} .

Before going any further, it is essential to introduce our finding named “the law of discrimination angles,” which is shown as below.

Law of Discrimination Angles: If there are two points on the circle O (denoted as points D and E respectively) whose discrimination angle constructed by the OneWeb satellite (point A), the point itself and the GSO satellite (point B) equals the threshold α_{th} , as seen in Figure 6, then for any point $P \in \widehat{DE}$ on the circle O , its corresponding discrimination angle is no smaller than α_{th} .

Above all, we give the proof of this law, based on two propositions about the circumferential angle listed as follows:²⁸

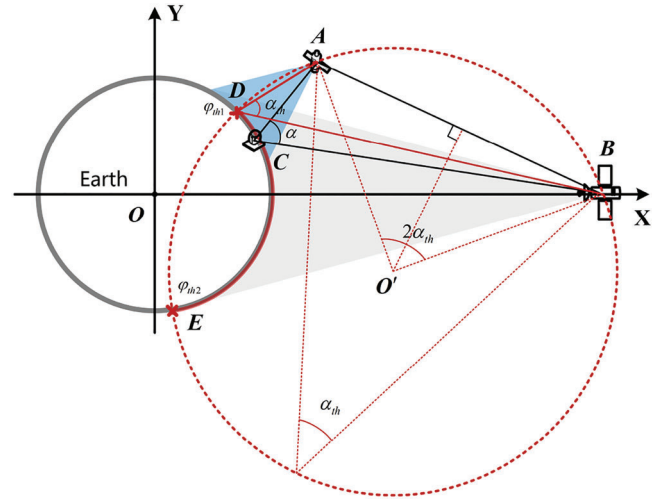
Proposition 1. In equal circles, the circumferential angles subtending the same chord equal one another, which are half the size of the corresponding central angle.

Proposition 2. If two unequal circles cut each other at the same chord, the circumferential angle subtending the chord in the smaller circle is greater than that in the larger circle.

In the field of geometry, these two propositions are easy to prove correct and can be used directly. Owing to $\angle ADB = \angle AEB$, we can find that points A, B, D , and E are concyclic in accordance with the Proposition 1. Besides, as illustrated in Figure 6, $\widehat{DE}$ is within the circle determined by the previously mentioned four points, so it is true that any point $P \in \widehat{DE}$ satisfies $\angle APB \geq \alpha_{th}$ on the basis of Proposition 2. In consequence, we have proved the correctness of the law of discrimination angles.

Then, we need to find out the points D and E in virtue of the law, so as to obtain the target coverage of the OneWeb satellite beam satisfying the discrimination angle. When the positions of the GSO satellite and OneWeb satellite are once determined, we can draw a circle with equal

FIGURE 6 Geometric diagram of analyzing the circle with equal discrimination angles



discrimination angles, whose chord is the line segment AB constructed by the GSO satellite and OneWeb satellite, and the corresponding central angle is $2\alpha_{th}$. This circle is centered at O' and cuts the circle O at the points D and E , whose radius is

$$R = \frac{|\overrightarrow{AB}|}{2\sin \alpha_{th}} = \frac{\sqrt{R_S^2 + R_G^2 - 2R_S R_G \cos \varphi_S}}{2\sin \alpha_{th}}. \quad (11)$$

Assume the coordinates of the circle center O' to be $(x_{O'}, y_{O'})$. Because O' is on the perpendicular bisector of the line segment AB , the relationship between $x_{O'}$ and $y_{O'}$ can be expressed as

$$y_{O'} = -\frac{R_S \cos \varphi_S - R_G}{R_S \sin \varphi_S} x_{O'} + \frac{R_S^2 \cos^2 \varphi_S - R_G^2}{2R_S \sin \varphi_S} + \frac{R_S \sin \varphi_S}{2}. \quad (12)$$

And point B is on the circle O' , thus

$$(x_{O'} - R_G)^2 + y_{O'}^2 = R^2. \quad (13)$$

Substituting (11) and (12) into (13), we obtain the position coordinates of the circle center O' as follows:

$$\begin{cases} x_{O'} = \frac{R_S \cos \varphi_S + R_G}{2} - \frac{R_S \sin \varphi_S}{2 \tan \alpha_{th}}, \\ y_{O'} = \frac{R_S \cos \varphi_S - R_G}{2 \tan \alpha_{th}} + \frac{R_S \sin \varphi_S}{2}. \end{cases} \quad (14)$$

Denote the latitudes of D and E as $\varphi_{th,1}$ and $\varphi_{th,2}$ respectively, then their coordinates can be expressed as $(R_E \cos \varphi_{th,1}, R_E \sin \varphi_{th,1})$ and $(R_E \cos \varphi_{th,2}, R_E \sin \varphi_{th,2})$, which satisfy the following equation:

$$(R_E \cos \varphi_{th,i} - x_{O'})^2 + (R_E \sin \varphi_{th,i} - y_{O'})^2 = R^2, \quad i = 1, 2. \quad (15)$$

Finally, solve the equations group composed of (14) and (15), we can get the analytical solutions of $\varphi_{th,1}$ and $\varphi_{th,2}$, which are the boundaries of OneWeb satellite beam' target coverage.

$$\begin{cases} \varphi_{th,1} = \arcsin \left(\frac{ab}{1+a^2} + \frac{\sqrt{(1+a^2)R_E^2 - b^2}}{1+a^2} \right), \\ \varphi_{th,2} = -180 - \arcsin \left(\frac{ab}{1+a^2} - \frac{\sqrt{(1+a^2)R_E^2 - b^2}}{1+a^2} \right). \end{cases} \quad (16)$$

where $a = \gamma_{O'} \cdot x_{O'}$ and $b = R_G + (R_E^2 - R_G^2) \cdot 2x_{O'}$.

On the basis of the previous analysis and these analytical solutions, we can further transform the discrimination angle constraint into two boundary constraints, and the primal optimization problem in (7) can be ultimately converted into the new optimization problem above.

$$\begin{aligned}
 & \min_{\theta, N_{OFF}} C(\theta, N_{OFF}) \\
 \text{s.t. } & C1_1'' : \varphi_{B1}(\theta, N_{OFF}) \leq \varphi_{th,1}, \\
 & C1_2'' : \varphi_{B2}(\theta) \geq \varphi_{th,2}, \\
 & C2 : 0 \leq \theta \leq \theta_{max}, \\
 & C3 : N_{OFF} \in \{0, 1, \dots, N_{ellipse}\}.
 \end{aligned} \tag{17}$$

3.2 | Solutions of optimization problem

As described in the previous section, the primal problem in (7) is transformed into the new optimization problem, and the analytical relationship between the boundary constraints $C1_1''$, $C1_2''$ and the optimization variables θ , N_{OFF} is clarified. Analyzing the definition of the implementation cost function, we can find that $C(\theta, 0) \leq C(\theta_{max}, 0) \leq C(\theta_{max}, N_{OFF})$, and equality holds exactly when $\theta = \theta_{max}$ and $N_{OFF} = 0$. Firstly, we need to judge the boundary constraints are satisfied or not when the equality condition is achieved. Then, according to the judgment, adjust the value setting of θ or N_{OFF} to solve the optimization problem. Specific ideas are as follows:

- 1) If the boundary constraints are satisfied when the equality condition is achieved, it means that adjusting OneWeb satellite's attitude can avoid harmful interference and there is no need to turn off beams. Based on this, let $N_{OFF} = 0$, and the problem in (17) is rewritten into a univariate optimization problem with respect to θ ,

(18)

In accordance with the analysis in the previous subsection, it is obvious that the constraint $C1_1''$ is stronger than $C1_2''$. Furthermore, when N_{OFF} is determined, φ_{B1} decreases along with the increase of θ . Hence, the optimal solution θ^* can be obtained at $\varphi_{B1}(\theta^*, 0) = \varphi_{th,1}$.

- 2) If the boundary constraints are not satisfied when the equality condition is achieved, it indicates that the maximum pitch angle is not enough for the OneWeb satellite to mitigate interference, and user beams need to be partially or completely shut down. We set $\theta = \theta_{max}$, and the problem in (17) is transformed into a univariate optimization problem with respect to N_{OFF} (N_{OFF} has a finite number of values),

$$\begin{aligned}
 & \min_{N_{OFF}} N_{OFF} \\
 \text{s.t. } & C1_1'' : \varphi_{B1}(\theta_{max}, N_{OFF}) \leq \varphi_{th,1}, \\
 & C3 : N_{OFF} \in \{1, \dots, N_{ellipse}\}.
 \end{aligned} \tag{19}$$

The constraint $C1_2''$ is independent of N_{OFF} , so we only need to consider the constraint $C1_1''$. When θ is determined, φ_{B1} decreases along with the increase of N_{OFF} . Therefore, the optimal solution N_{OFF}^* ought to be obtained at $\varphi_{B1}(\theta_{max}, N_{OFF}^*) = \varphi_{th,1}$.

Combining the above two cases, we can obtain the optimal solution of the optimization problem in (17),

$$\begin{cases} \theta^* = \min(\theta_{max}, \beta_0 - \arctanc), \\ N_{OFF}^* = \left\lceil \frac{\beta_0 - \theta^* - \arctanc}{2\beta_{ellipse}} \right\rceil. \end{cases} \tag{20}$$

where $c = \frac{R_E \sin(\varphi_{th,1} - \varphi_S)}{R_S - R_E \cos(\varphi_{th,1} - \varphi_S)}$ and $\lceil \cdot \rceil$ means rounding a number upwards to the nearest integer.

After obtaining the optimal value of θ^* and N_{OFF}^* , we also need to determine the sign of θ^* . Note that the sign of OneWeb satellite's pitch angle is related to both the motion state and the position of the satellite. Consider the case where OneWeb satellite crosses the equator at its ascending node, i.e. from south to north. If OneWeb satellite is over the northern hemisphere, it needs to be tilted towards south, while θ^* is negative. If OneWeb satellite is above the Southern Hemisphere and tilt itself towards north, θ^* ought to be positive, whereas the sign of θ^* in the case of OneWeb satellite passing over the equator from north to south at its descending node is exactly opposite to that of the ascending node case.

Thus far, the primal optimization problem in Equation (7) is solved accordingly, and we can acquire a set of progressive pitch parameters corresponding to the OneWeb satellite at different latitudes through the above method.

Due to the symmetries of uplink and downlink, the derivations are similar, but in uplink case, the interference source is the earth station, and the interference to the GSO satellite should not exceed the threshold, which means Equation (5) should be rewritten as

$$I = P_{ES,T'} \cdot G_{RS} \cdot G_{TG}(\alpha(\varphi_{ES})) \cdot \left(\frac{\lambda}{4\pi d_{ES-S}} \right)^2, \quad (21)$$

where $P_{ES,T'}$ represents the power of earth station's transmitter in the GSO network's operating band and G_{RS} denotes the maximum receiving gain of GSO satellite. $G_{TG}(\alpha(\varphi_{ES}))$ represents the transmit antenna gain of earth station in the off-axis angle $\alpha(\varphi_{ES})$ direction. Another difference is in the calculation of α_{th} in Equation (9). The threshold value of discrimination angle corresponding to the interference threshold can be redefined as follows:

$$\alpha_{th} = G_{TG}^{-1} \left(\frac{I_{th} \cdot (4\pi d_{min})^2}{P_{ES,T'} \cdot G_{RS} \cdot \lambda^2} \right). \quad (22)$$

4 | NUMERICAL RESULTS

4.1 | Acquisition of progressive pitch's parameters

First of all, we need to obtain the exact values of the progressive pitch's parameters, as the input of the subsequent numerical interference analysis.

Specifically, we analyze the Ku-band spectral coexistence as described in Section 2. Based on the radio regulations of ITU, the interference protection criteria is selected that $I \leq 0.06N_G$, where N_G means the noise level at the receiver of GSO satellite network.²⁹ In other words, the interference-to-noise-ratio (INR) at the GSO earth station's receiver shall not exceed -12.2 dB. As for the communication parameters, we refer to CHNNEWSAT-G1-118E's network database published by ITU,³⁰ as well as OneWeb's FCC technical fillings (see next subsection for details). Then, substitute both the interference protection threshold value and aforementioned parameters into the optimization problem in Section 2, and solve it on the strength of the method in Section 3, and we can acquire the overall progressive pitch's parameters (including the mapping relationship between the latitude of OneWeb's subsatellite point, the exact number of beams to be shut off and the pitch angle of OneWeb satellite), as seen in Figure 7.

Figure 7 indicates the case where the OneWeb satellite passes over the equator from south to north. It shows that both the pitch angle θ and the number of shut-off beams N_{OFF} are symmetric about $\varphi_S = 0$; that is, the progressive pitch operations of OneWeb satellite in the Northern Hemisphere and Southern Hemisphere are mirror symmetric. In detail, when the latitude of OneWeb satellite is greater than 30° , there is no need to adjust attitude of OneWeb satellite and to turn off beams for interference mitigation, with both θ and N_{OFF} being zero. Then, the absolute value of pitch angle increases monotonically with decrease of φ_S until φ_S equals 21.96° . When the OneWeb satellite is above the areas within the

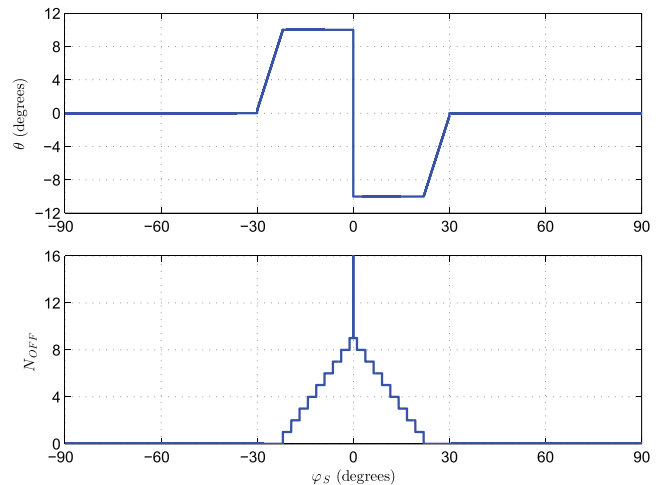


FIGURE 7 The mapping curve of the progressive pitch's parameters varying with the latitude of OneWeb's subsatellite point φ_S

latitude of 21.96° , θ exactly equals θ_{max} as 10° , and the number of shut-off beams has a ladder-like increase as the latitude of OneWeb satellite decreases. Besides, when OneWeb satellite passes over the equator, i.e., $\varphi_s = 0$, all the beams need to be turned off momentarily while satellite rotating to the symmetrical direction.

4.2 | Interference analysis

In this subsection, numerical simulation results are provided to verify the interference avoidance effect of OneWeb system's progressive pitch strategy with the previously obtained parameters. The simulation is made with the actual orbital configuration and communication parameters of both CHNNEWSAT-G1-118E satellite network and OneWeb system (listed in Tables 1 and 2). To establish the simulation model, we select only one orbital plane of OneWeb system, with a total of 40 satellites, while the GSO satellite network named CHNNEWSAT-G1-118E is located at longitude of 8°E with orbital inclination of 0° . In the meantime, we choose four locations at different latitudes to set up GSO earth stations. These four earth stations have the same longitude of 8°E as the GSO satellite, nevertheless varying in latitudes of 5°N , 20°N , 40°N , and 60°N , respectively. They are all within the coverage areas of both OneWeb system and GSO satellite, whose main lobe of antenna keeps pointing to the direction of the GSO satellite.

During the simulation process, OneWeb satellites are tilted, combined with turning off certain beams, according to the parameters of progressive pitch strategy acquired in the previous subsection. Besides, we take the case that OneWeb satellites are always oriented to the earth without attitude adjustment and beams turning-off for comparison, and we can obtain the following simulation results.

Figures 8 and 9 present the INR values of GSO earth station when it is located at latitudes of 5°N and 20°N . Additionally, for case that the interfering OneWeb satellites adopt the progressive pitch strategy, its pitch angle and exact number of the turned-off beams are also shown at the bottom of these two figures. We can find that when the progressive pitch strategy with the previously obtained parameters is applied to OneWeb system, the INR values are significantly reduced below the -12.2 dB threshold. However, blind areas of OneWeb's service appear in some time with the progressive pitch strategy. We take the first three minutes in Figure 8 as an example to carry on concrete analysis. In the closeup view, the first piece of red curve represents the case for comparison that the GSO earth station is interfered by OneWeb Satellite #N, and then the interferer turns to be the postpositional Satellite #N-1. At the same time, in the "progressive pitch" mode, the GSO earth station constantly receives the signal from the satellite at higher latitude, compared with former case. It is due to OneWeb satellite's beams tilting towards the equator. As for the blue curve in the enlarged Figure 8, we can see that the area where the GSO earth station is located in is served by OneWeb Satellite #N+1, #N-1, and #N in turn, and no service is available in this area for a short time before user can switch to Satellite #N-1. The whole progress is illustrated in Figure 10. In detail, in Figure 10a, earth station is serviced by Satellites #N+1, and in Figure 10b, Satellites #N turns off all the beams to mitigate the interference, causing no service on earth station. Later, the earth station is in the service area of Satellite #N for a short period of time, for Satellite #N turns south and Satellite #N-1 turns north. Finally, the earth station is serviced by Satellite #N and start the same cycle again.

Parameter	Value
Downlink carrier frequency	12.7 GHz
Downlink bandwidth	1.28 MHz
Diameter of GSO earth station's receive antenna	0.6 m
Peak gain of GSO earth station's receive antenna	35.7 dBW
Beamwidth of GSO earth station's receive antenna	2.8°
Antenna pattern of GSO earth station	REC-465-5
Equivalent system noise temperature	150 K

Abbreviation: GSO, geostationary orbit.

TABLE 1 GSO network's Ku-band downlink communication parameters

Parameter	Value
Downlink carrier frequency	12.575 GHz
Downlink bandwidth	250 MHz
Satellite EIRP	34.6 dBW
Satellite PSD	-13.4 dBW/4 kHz

Abbreviation: EIRP, equivalent isotropically radiated power; PSD, power spectral density.

TABLE 2 OneWeb system's Ku-band downlink communication parameters

FIGURE 8 Interference to noise ratio (INR) of the geostationary orbit (GSO) earth station at latitude of 5°N with the pitch angle and exact number of the turned-off beams of the interfering OneWeb satellite

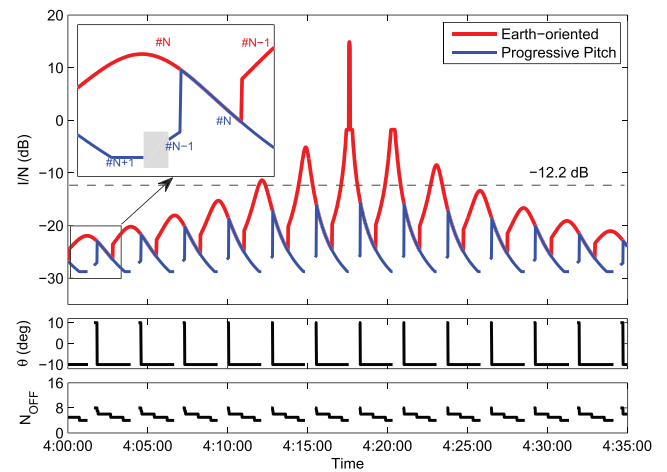


FIGURE 9 Interference to noise ratio (INR) of the geostationary orbit (GSO) earth station at latitude of 20°N with the pitch angle and exact number of the turned-off beams of the interfering OneWeb satellite

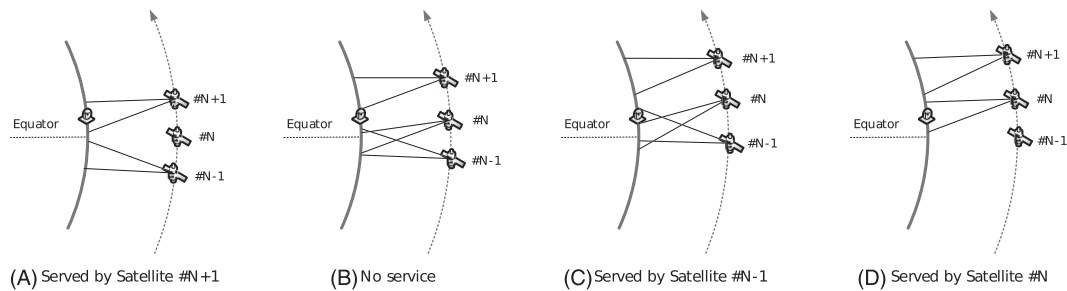
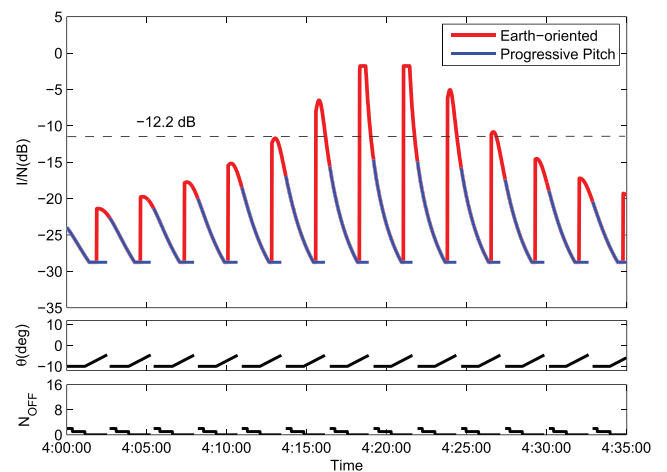


FIGURE 10 The process of an earth station at latitude 5°N served by the OneWeb system when the progressive pitch strategy is adopted

Besides, in accordance with previous analysis, when OneWeb satellite passes over the areas of middle and high latitude (greater than 30° , as shown in Figure 7), there is no need to adjust its attitude and even turn off the user beams. As shown in Figures 11 and 12, simulation results verify that whether adopting the progressive pitch strategy or not is essentially the same for these areas, where the INR values remain below the interference threshold.

5 | ANALYSIS OF BLIND AREAS

In this section, we mainly analyze the causes of blind areas based on the previous mathematical model. In OneWeb constellation, each satellite is discretely distributed along the LEO, and the pitch degree of each satellite varies with its latitude, which makes the overlap coverage of adjacent

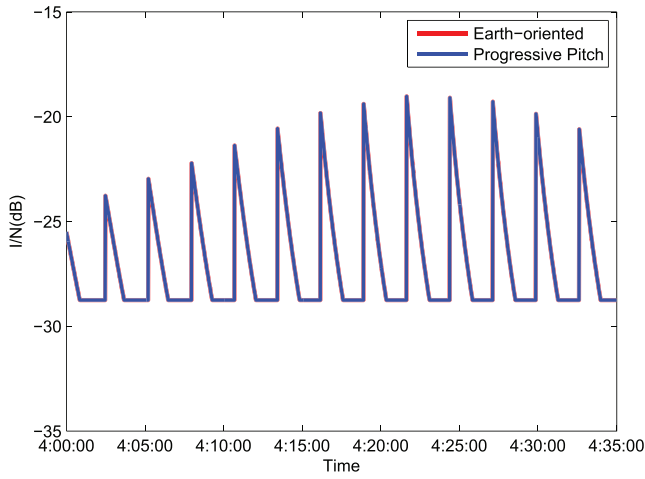


FIGURE 11 Interference to noise ratio (INR) of the geostationary orbit (GSO) earth station at latitude of 40°N

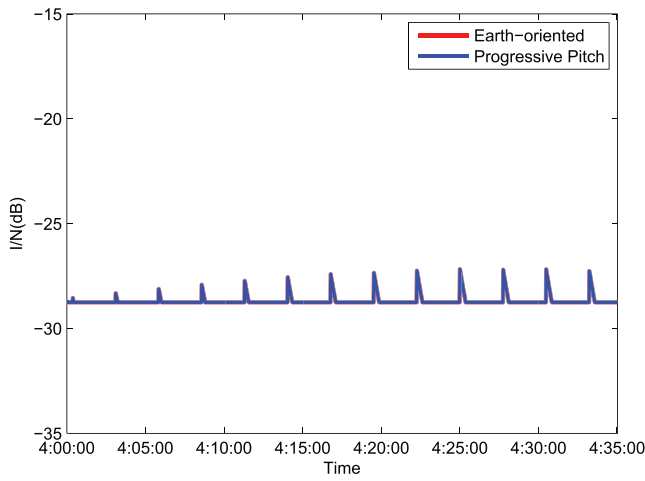


FIGURE 12 INR of the GSO earth station at latitude of 60°N

satellites in the same orbit smaller. Besides, as stated before, the satellites above low-latitude areas need to shut off several beams. Both of the above reasons lead to the fact that the progressive pitch strategy may bring about blind areas for service in some time while avoiding interference. Specifically, when the OneWeb satellite is over the equator, its all beams need to be turned off, and the front and rear satellites are not tilted enough so that their beams cannot cover the equatorial region, making it impossible to provide communication services to that region.

We denote the OneWeb satellite with latitude of φ_S as #N. Assume that the phase difference between adjacent OneWeb satellites is δ , and then the subsatellite point's latitude of the prior satellite marked as #N+1 should be $\varphi_S + \delta$. According to Equation (2) in Section 2, under the progressive pitch strategy, the upper edge of Satellite #N's beam coverage can be expressed as

$$\varphi_{B1_N} = \varphi_S + \arccos(g(\beta_N - \theta_N)). \quad (23)$$

And the lower edge of Satellite #N+1's beam coverage can be denoted as

$$\varphi_{B2_{N+1}} = \varphi_S + \delta - \arccos(g(\beta_0 + \theta_{N+1})). \quad (24)$$

Thus, the overlap coverage of these two adjacent satellites can be presented as follows:

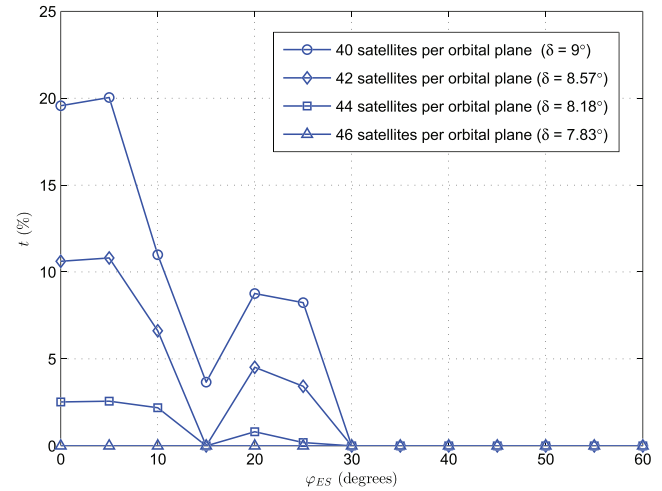
$$\begin{aligned} \Delta\varphi_B &= \varphi_{B1_N} - \varphi_{B2_{N+1}} \\ &= \arccos(g(\beta_N - \theta_N)) + \arccos(g(\beta_0 + \theta_{N+1})) - \delta. \end{aligned} \quad (25)$$

If $\Delta\varphi_B \geq 0$, it means that there is an overlap between adjacent satellites' coverage, which provides dual coverage. Although $\Delta\varphi_B < 0$, blind areas for service appear. To eliminate the blind areas, we need to make $\Delta\varphi_B$ equal or greater than 0. In terms of the expression in Section 2, it is

TABLE 3 The effect of blind areas elimination method on variables corresponding to $\Delta\varphi_B$

Method index	Variables					$\Delta\varphi_B$
	δ	θ_N	θ_{N+1}	β_N	β_0	
1	↓	↓/—	—/↑	—	—	↑
2	—	↓	↑	—	—	↑
3	—	—	—	↑	—	↑
4	—	—	—	—	↑	↑

FIGURE 13 The percentage of no-service time at all latitudes on the ground when the number of OneWeb satellites per orbit is 40, 42, 44, and 46



easy to find that $g(\cdot)$ is a monotone decreasing function within the interval $\left[0, \arcsin\left(\frac{R_E}{R_S}\right)\right]$. Therefore, in accordance with Equation (24), we can find the following ways to reduce or even eliminate the blind areas (the effects on variables corresponding to $\Delta\varphi_B$ are listed in Table 3).

- Method 1: Adding satellites per orbital plane.
- Method 2: Increasing the tilt of Satellite #N+1 at higher latitude, and decreasing the tilt of Satellite #N at lower latitude.
- Method 3: Reducing the number of Satellite #N shut-off beams to enlarge the horizontal half angle of its equivalent rectangular beam.
- Method 4: Enlarging the original horizontal half angle of each satellite's equivalent rectangular beam.

Hereon, we briefly analyze the effect of adding satellites per orbital plane on eliminating blind areas through numerical simulation. Figure 13 presents the percentage of no-service time at all latitudes on the ground when the number of OneWeb satellites per orbit is 40, 42, 44, and 46. It can be observed that the percentage of no-service time at all latitudes on the ground overall decrease as the number of satellites increases. When the satellites in one orbital plane grows to 46, the no-service time ratios at all latitudes are all reduced to zero, which indicates that adding six satellites per OneWeb's orbit can eliminate blind areas worldwide when the progressive pitch strategy is adopted.

6 | CONCLUSIONS

In this paper, we focus on the parameters acquisition process and the interference avoidance effect of OneWeb's progressive pitch strategy. A mathematical system model of the spectral coexistence scenario between OneWeb system and a GSO satellite network is established to formulate an optimization problem according to the technical route of the progressive pitch strategy, which is constrained by the interference protection threshold of GSO satellite networks. We do a series of transformations with the geometrical analysis method to simplify this optimization problem and finally obtain the progressive pitch's parameters suitable for OneWeb system. Furthermore, we present the cofrequency compatibility simulation results between OneWeb system and GSO satellite system in the global scope. The numerical simulation results verify that, progressive pitch strategy with the matchable parameters obtained by our proposed method can enable OneWeb system to avoid harmful influence to the GSO satellite network in most parts of the world, with less loss of its service capability.

In general, the performance of progressive pitch strategy is quite in line with the initial planning objective of OneWeb constellation. However, blind areas with no service may appear in some time, when the progressive pitch strategy is adopted. It is the inevitable result of the combined

effect of the discrete distribution of LEO satellites, the tilting difference of adjacent satellites in one orbit, and the shutting off of partial beams. In the further research, we can explore more effective ways to reduce or even eliminate the blind areas in depth.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Ting Li  <https://orcid.org/0000-0002-5417-030X>

Jin Jin  <https://orcid.org/0000-0002-8500-5016>

Zixuan Ren  <https://orcid.org/0000-0002-9141-5919>

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AUTHOR BIOGRAPHIES



Ting Li received her BS degree from Beijing University of Posts and Telecommunications, Beijing, China, in 2017. She is currently pursuing the MS degree in School of Aerospace Engineering, Tsinghua University. Her research interests lie in the satellite communications and cofrequency interference mitigation.



Jin Jin received his PhD degree in Navigation, Guidance and Control from Beijing Institute of Technology, China, in 2011. After 2 years as a post-doctoral research assistant in Tsinghua University, he joined Tsinghua Space Center, Tsinghua University as an assistant professor in 2013. Currently, he is an associate researcher in Beijing National Research Center for Information Science and Technology. His research interests are space mission design and space communication system.



Wei Li received his PhD degree from Beijing University of Posts and Telecommunications (BUPT) in 2013. Now, he is a senior engineer in the satellite frequency and orbit resource agency, the State Radio Monitoring Center of China. His research interests include intersystem frequency compatibility analysis and interference management for space communication systems.



Zixuan Ren received his BS degree in electronic engineering from Tsinghua University, Beijing, China, in 2016. He is currently pursuing his PhD degree with School of Aerospace Engineering, Tsinghua University. His research interests include cofrequency interference mitigation and satellite communications.



Linling Kuang received her BS and MS degrees from National University of Defense Technology, Changsha, China, in 1995 and 1998, respectively, and PhD degree in electronic engineering from Tsinghua University, Beijing, China, in 2004. Since 2007, she has been with Tsinghua University, where she is currently a professor in Beijing National Research Center for Information Science and Technology. Her research interests include wireless broadband communications, signal processing, and satellite communication.

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